Toheeb Babatunde Ibrahim

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RESEARCH INTERESTS

- Mathematical Biology
- Mathematical Modeling
- Numerical Analysis
- Physics Informed Neural Network
- Control Systems
- Quantitative Analysis

EDUCATION

Bowling Green State University **Aug. 2024 – Present**
Doctor of Philosophy in Applied Mathematics *Bowling Green, Ohio, US*

Gdańsk University of Technology **Oct. 2023 – Jan. 2025**
Master of Science in Nanotechnology (Dóbry Plus) *Gdańsk, Poland*

University of L'Aquila **Sep. 2022 – Sep. 2023**
Master of Science in Mathematical Engineering (108/110) *L'Aquila, Italy*

African Institute for Mathematical Sciences **Oct. 2018 – Jun. 2019**
Master of Mathematical Sciences (Very Good Pass) *Limbe, Cameroon*

University of Ilorin **Sep. 2013 – Oct. 2017**
Bachelor of Science in Mathematics (First Class Honors) *Ilorin, Nigeria*

RELEVANT COURSEWORK

- Discrete & Continuum Mechanics
- Matrix Analysis
- Mathematical Analysis
- Control Systems
- Physics of Materials
- Methodology of Scientific Research
- Convex Optimization
- Computer Modeling & Design of materials
- Probability theory

RESEARCH EXPERIENCES

Hybrid Neural ODE-Based Dynamic Modeling and Observer-Based Material-Balance Regulation of Industrial Glass Melters **2026**

College of Engineering, Bowling Green State University

Authors: Owais Khan, Xuan Dung James Nguyen, Toheeb Babatunde Ibrahim, Mohammed Abouheaf, Josh Peeno

- Developed a first-principles model of the glass melting process dynamics.
- Developed a Luenberger observer that estimates the unmeasured process states needed for LQR setup.
- Applied the NI9 method to solve problems in epidemiology and non-linear logistic model.

Stochastic Divergence in Mortality and the effect on Social Security's Progressivity **2026**

Bowling Green State University

Toheeb Babatunde Ibrahim, Patrick Nyadjo

- Conducted an in-depth exploration of fundamental concepts social security.
- Investigated different actuarial models for social security benefits.
- Explored stochastic model for economic-dependent mortality and its implication on social benefit calculation.

Finding the Central-Force Decomposition of the 2NN-MEAM Potential **2024**

Gdańsk University of Technology

Advisor: Dr Szymon Winczewski

- Carried out in partial fulfilment for an award of Master's degree in Nanotechnology
- Gave an extensive description of the Sturm-Liouville problem by defining the problem together with the notion of Self-adjoint differential operators.

WORK EXPERIENCE

College of Engineering, Bowling Green State University **August 2025 – Present**
Research Assistant *Bowling Green, Ohio, US*

- Develop a physics-based control system model for an industrial glass melting furnace
- Analysed control architecture including PID, LQR and observer-based control.
- Implemented advanced controller tuning technique, including PSO-tuned PID/FOPID.
- Study the mathematical theory in the model reduction of CFD models to simpler mathematical/control model
- Organize and cordinating meeting schedules with industrial patners.

Department of Mathematics & Statistics, Bowling Green State University **August 2024 – May 2025**
Graduate Teaching Associate *Bowling Green, Ohio, US*

- Taught first-year students College algebra in the Math Emporium.
- Taught Calculus and Analytical Geometry to Biology and Pre-Med students
- Cordinate the course and implemented the course curriculum to the student.
- Implemented a computer assisted teaching e.g. Geogebra, Desmos etc for visualization of mathematical concepts.

African Institute for Mathematical Science **Nov. 2019 – Dec 2019**
Teaching Assistant *Limbe, Cameroon*

- Assisted in teaching Mathematical Modeling of Dynamical Systems by providing tutorial for the student of the course
- Guided the student in the use of Berkeley Madonna for the simulation of their dynamical systems model.
- Guided the student on the interpretation of model and translating their result into policy.

Adekunle Ajasin University **Dec. 2017 – Oct. 2018**
Teaching Assistant *Akungba-Akoko, Nigeria*

- Tutored First-year students in the departments of mathematics on foundational mathematics.
- Assisted final year students with MATLAB simulation

SCHOLARSHIPS AND AWARDS

J. Robert & G. Overman Scholarship Award **April 2025**

Recipient of the award and scholarship for a graduate student of Mathematics and Physics nominated by the faculty of the department in Bowling Green State University.

Bowling Green State University Graduate Assistanship **August 2024**

Recipient of the fully funded Doctor of Philosophy program in Applied Mathematics at Bowling Green State University for the 2024/2025 till date.

Erasmus+ Scholarship **September 2022–2024**

Recipient of the Erasmus+ scholarship for high-performing students to support their studies in their parent university and their host university during exchange program.

Abruzzo Regional Scholarship **September 2023–2024**

Recipient of the Abruzzo regional scholarship for student that graduated within the stipulated time of their program and with GPA above 105/110

University of L'Aquila International Student Scholarship **September 2022–2023**

Recipient of the International Student Scholarship for student that ranked in the first ten during their application.

African Institute for Mathematical Science **October 2018–2019**

Recipient of the fully funded Master of Science in Mathematical Sciences at African Institute for Mathematical Sciences (AIMS) Cameroon for the 2018/2019 Academic Session.

DAAD Scholarship **September 2019–June 2021**

Sponsored to University of Tuebingen and Hochschule Mittweida University of Applied Sciences as a guest researcher to work as a research assistant under the mentorship of a German-based professor.

Petroleum Technology Development Fund **April 2016**

Recipient of the scholarship award to outstanding undergraduate students in the Sciences and Mathematics in Nigeria universities.

National Mathematical Center, Abuja **April 2016**

A Gold Medal for distinction among the nominees of each Nigeria universities at the National Mathematics Competition among University Students.

PUBLICATION

Kristina B.H., Girma M.Z, **Toheeb B.I.**, Aliou B., Henri T.O., Vicent A., Gideon A. N., Miranda T. E, Kristan A.S. *Is increased mortality by multiple exposures to COVID-19 an overseen factor when aiming for herd immunity?*. PLOS One (2021). Link

Nessma A.Y., Henri T.O., Yvan Jordan N.M., Sandy K.N., Loyce K., George K., **Toheeb B.I.**, Patience B.I., Sulyman I., Looli A.N., Kristina B.H., Miranda T.E. Kristan A.S. *The impact of COVID-19 vaccination campaigns accounting for antibody-dependent enhancement*. PLOS One (2021).Link

PRESENTATIONS

Toheeb B. Ibrahim, Sustaining efficient malaria vector control while accounting for breeding site stages: a predictive modelling approach – Presented at American Society of Tropical Medicine annual meeting 2021.

Toheeb B. Ibrahim, Mathemetical modeling of *Anopheles* mosquitoes' breeding sites to sustain efficient vector control- Presented at American Society of Tropical Medicine annual meeting 2020.

Toheeb B. Ibrahim, Is increased mortality by multiple exposures to COVID-19 an overseen factor when aiming for herd immunity? - Presented at Mathematical Epidemiology Conference hosted by the Center for Disease Modeling at York University 2021.

PROFESSIONAL DEVELOPMENT TRAININGS

AI-ready Ohio **2026**

Enterprise Technology Association *Ohio*

- Training on the use of AI in a professional environment.
- Training on the use of AI for policy makers.

Great Lakes AI week Conference **2025**

Bowling Green State University *Ohio*

LEADERSHIP EXPERIENCES

Muslim Student Association **August 2025 – Present**

Vice President *Bowling Green State University, Ohio*

- Ensured regular compliance of the association activities with the university guidelines
- Coordinated the association activities, including outreach and publicity to the freshmen
- Correspondence with faculty matrons.

Nanotechnology Erasmus Students Lab Work **October 2023 – November 2023**

Project Lead *Gdańsk, Poland*

- Coordinate laboratory schedule and booking for use among the nanotechnology students.
- Coordinated laboratory activities and took inventory of laboratory materials being used.
- Raised any concerns regarding laboratory equipment to the course coordinator.

VOLUNTARY EXPERIENCES

Sustainable Development Goals, National Youth Service **Dec. 2017 – Oct. 2018**

Vice President *Ondo State, Nigeria*

- Mobilize funds and training resources needed to give vocational training to the residents of the rural areas of the town– targeting Poverty Eradication of the SDGs
- Organized sensitization on drug use among adults of the community– targeting Good Health of the SDGs.
- Organized quiz competition among high school students– targeting Education of the SDGs.

SCHOLARLY MEMBERSHIP

Member, American Mathematical Society **August 2024 - Present**

TECHNICAL SKILLS

Language: Python, MATLAB, *R*

Data Analytic: MS Excel, SQL, GNUPlot

Analytical: Research, Report Writing, LaTeX Typesetting, Creative Writing

Information Communication Technology: Microsoft Office Suite (MS Word, MS Excel, and Powerpoint)

LANGUAGES

Enlish: Full professional proficiency – Distinguished levels in Listening, Speaking, Reading and Writing.

Italian: A2 Level

French: A2

Polish: A1

Yoruba: Native proficiency

ADDITIONAL INFORMATION

- Good team player with the ability to working in a multidisciplinary team.
- Good time management with precise attention to details.
- Ability to guide young minds with different mindset and backgound.
- Flexible, quick to learn, highly organized and resourceful

REFERENCES

Available upon request